

Mahmoud Alyosify

Specialised Edge & Efficient AI Engineer | Inference Optimisation, Latency & Energy-Aware Deployment | MSc Artificial Intelligence, Queen's University (Canada)

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🌐 [linkedin.com/in/mahmoudalyosify](https://www.linkedin.com/in/mahmoudalyosify) | 🐙 github.com/MahmoudAlyosify | 📁 Portfolio | 🤖 Hugging Face | 🏆 Kaggle

PROFESSIONAL SUMMARY

Efficient-AI Engineer whose entire focus is the constraint most ML work ignores: **making models cheap enough to actually run**. Master's research at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) delivers a **training-free, zero-overhead decoding method** that cuts generated tokens across three model families while measuring **TTFT, inter-token latency and GPU energy through NVML** — efficiency treated as a measured result, not a claim. Practical deployment experience with **ONNX export and FAISS-backed inference, 4-bit quantised fine-tuning (QLoRA), self-hosted model serving, and deep-RL resource provisioning under cost-latency budgets**. Systems background in **C/C++ and real-time national toll infrastructure**, plus a **200-hour ML internship with ITIDA / Egypt Makes Electronics**. Seeking edge, on-device and inference-optimisation engineering roles.

EDUCATION

- MSc in Artificial Intelligence**
Queen's University, Kingston, Ontario, Canada

Sep 2025 – Nov 2026
Expected Nov 2026

 - **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
 - **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
 - Coursework: **Large Language Models, Deep Learning, Cloud Computing, Reinforcement Learning, End-to-End ML Pipelines.**
- B.Sc. Computer & Information Science (Bioinformatics)**
Aassiut University, Egypt

Sep 2019 – Jul 2023
GPA: 3.53 / 4.00

 - Graduation Project: **Vitalism Solution** — real-time contactless vital-sign estimation from video (rPPG). **Grade A+; Semi-Finalist, Microsoft Imagine Cup 2023.**

TECHNICAL SKILLS

Inference Optimisation	Decoding-Time Token Reduction, Custom LogitsProcessors, Entropy-Based Gating, KV-Cache-Free Methods, Quantised Fine-Tuning (QLoRA / 4-bit), PEFT / LoRA Adapters, Self-Hosted Model Serving
Profiling & Measurement	Time-to-First-Token (TTFT), Inter-Token Latency, Throughput Benchmarking, GPU Energy Measurement via NVML , Deterministic Benchmarking on Pinned Hardware, Cost-Latency Trade-off Analysis
Deployment Runtimes	ONNX, FAISS Vector Search, Docker, FastAPI, Streamlit, AWS GPU Instances, Linux, CI/CD
ML & Deep Learning	PyTorch, TensorFlow, Hugging Face Transformers, CNNs, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Scikit-learn, Optuna
Systems & Low-Level	C, C++, Real-Time Application Support, File-System Implementation (Mini-FAT), Embedded-Oriented ML Training (ITIDA / Egypt Makes Electronics, 200 hrs), Git/GitHub, Bash
Programming	Python, C, C++, C#, SQL, Go, MATLAB, Bash

SELECTED EFFICIENCY & DEPLOYMENT PROJECTS

- Verbosity-Aware Decoding — Zero-Overhead Token Reduction at Inference Time**
MSc Research Project (CISC 898), Queen's University · PyTorch · NVML · AWS GPU

2026 – Present

 - Built a **training-free LogitsProcessor** that cuts generated tokens at **decoding time** — **no retraining, no architecture change, no extra KV-cache or attention cost** — fusing stop-token, n-gram-repetition and semantic-redundancy signals under an **entropy-driven gate**.
 - Instrumented the full efficiency picture rather than token count alone: **TTFT, inter-token latency and GPU energy draw via NVML**, all captured under deterministic runs on **pinned AWS GPU instances**.
 - Proved the savings are not bought with quality loss, using **paired non-inferiority testing** (BERTScore, ROUGE, exact-match/F1, length-controlled LLM judge) across **Mistral-7B, Llama-3.1-8B and Qwen2.5-3B** on MT-Bench, XSum and SQuAD-v2.
- SimCLR-Vision-SSL — From Trained Encoder to ONNX Retrieval Service**
PyTorch · SimCLR · SupCon · ONNX · FAISS

2026

 - Exported a self-supervised vision encoder to **ONNX** and served its embeddings through a **FAISS** index, turning a research model into a runnable, framework-independent retrieval component.
 - Reached **84.30% top-1 accuracy** after a **42-experiment augmentation sweep** plus a semi-supervised **SupCon** extension.
- RL-Cloud-Autoscaler — Learned Provisioning Under a Cost-Latency Budget**
PPO · DQN · Custom Gymnasium Environment · Resource Optimisation

2026

 - Built a custom simulation environment and trained **PPO** and **DQN** agents to allocate compute under an explicit **cost-latency reward**, benchmarked against static and threshold-based policies.

- Fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on **AWS** for self-hosted interactive inference — keeping model weights and data in-environment rather than calling a hosted API.

Additional Projects

- **AudioShield** — compact **CNN** over mel-spectrograms delivering real-time audio classification through a lightweight served application.
- **Vitalism Solution** — real-time video signal processing (rPPG + wavelet filtering) for contactless vital-sign estimation from a standard camera.
- **Also:** Mini-FAT File System (C, low-level systems) · HORUS Sentinel (multi-agent, self-hosted LLM) · SCRAP · Automated Multimodal Agent.

PROFESSIONAL EXPERIENCE

Systems Engineer (Military Service)

National Company for Roads Building & Development

Feb 2024 – Mar 2025
Cairo, Egypt

- Supported **real-time backend systems** for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where latency and uptime were operational requirements, not metrics on a dashboard.
- Debugged and optimised live transportation software and supported legacy applications to minimise downtime; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

Machine Learning Instructor

National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025
Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.

AI in Cybersecurity Intern

e& Egypt — via NTI (On-the-Job Training)

Jun 2025 – Jul 2025
Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Online Instructor — AI, Algorithms & Data Science

Udemy & YouTube

2020 – Present
Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)	AWS Academy — ML Foundations, ML for NLP, Data Engineering
Machine Learning Specialization — DeepLearning.AI	HCIP – AI (Huawei)
Digilians Presidential Scholarship — Queen’s University	1st Place — Smart Cities Hackathon 2022
Semi-Finalist — Microsoft Imagine Cup 2023	Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.